

BAIRE GOWDA

Full Stack Developer | GenAI Engineer | Software Engineer

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CAREER OBJECTIVE

Motivated MCA student specializing in Cybersecurity and Machine Learning, with hands-on experience in full stack web development, Python-based AI applications, and REST APIs. Seeking a software developer or full stack developer role where I can apply my technical skills, deliver production-ready solutions, and grow within an agile engineering environment.

TECHNICAL SKILLS

Languages: Python, Java (Basics), JavaScript, HTML, CSS

Frontend: HTML5, CSS3, JavaScript

Backend: Python, Node.js (Basics), REST APIs

Databases: MySQL, SQL

AI / ML: Machine Learning, RAG Pipelines, LLMs, Prompt Engineering

Tools: Git, GitHub, VS Code, Streamlit, Postman, Streamlit Cloud

Concepts: Full Stack Development, OOP, Data Preprocessing, Cybersecurity Fundamentals

EDUCATION

Master of Computer Applications (MCA) — Chanakya University

2025 – 2027 | Pursuing

Specialization: Cybersecurity & Machine Learning

Bachelor of Computer Applications (BCA) — East Point College 2022 – 2025 | CGPA: 7.8

ACADEMIC PROJECTS

DocuMind AI — AI-Powered Knowledge Assistant

April 2026

Tech Stack: Python · Streamlit · RAG · LLaMA 3 · Sentence Transformers · OpenRouter API

- Built a production-ready RAG system that answers queries strictly from uploaded documents, eliminating AI hallucinations.
- Implemented a 5-stage pipeline: document ingestion, chunking, vector embedding, semantic retrieval, and grounded response generation.
- Deployed live on Streamlit Cloud with secure API key management and a custom NumPy-based vector store.

Enterprise Research Assistant (ERA)

June 2026

Tech Stack: React.js · FastAPI · Python · ChromaDB · FlashRank · GPT-4o-mini · OpenRouter API

- Built an enterprise-grade RAG application for intelligent document search and automated research report generation.
- Implemented semantic search, FlashRank reranking, Chain-of-Thought prompting, CRUD operations, and PDF report generation.
- Added security guardrails and API fallback mechanism for reliable and secure AI workflows.

Smart Parking System (IoT)

July 2026

Tech Stack: ESP32 · Arduino IDE · Blynk IoT · HC-SR04

- Developed an IoT-based Smart Parking System for real-time parking slot monitoring.
- Integrated ESP32, ultrasonic sensors, LEDs, and Blynk for vehicle detection and remote monitoring.
- Built a low-cost smart city parking prototype using Wi-Fi-enabled IoT technology.

SafeWalk — Student Safety Web Application

Dec 2025

Tech Stack: FlutterFlow, Base44

- Developed a student safety web application with live location tracking and SOS alerts.
- Implemented safe route guidance and an admin dashboard for real-time monitoring.
- Built a functional prototype using FlutterFlow and Base44 as part of an MCA Innovation Design Thinking project.

Heart Disease Prediction System

May 2025

Tech Stack: Python · Scikit-learn · Pandas · NumPy

- Developed an ML-based system to predict heart disease risk using real-world medical datasets.
- Performed data preprocessing, feature analysis, and model evaluation to improve prediction accuracy.
- Gained practical experience in end-to-end model development and data-driven problem solving.

Social Media Application

Jan 2025 – Feb 2025

Tech Stack: HTML · CSS · JavaScript · Python (Backend)

- Designed and developed a full-stack social media web application during a 90-hour internship.
- Implemented responsive frontend UI and integrated a Python-powered backend with core social features.

PUBLICATIONS

An Extended Study on High-Level K-Nearest Neighbors (HLKNN)

Zenodo (DOI Published) | July 2026

- Published a research paper on the HLKNN algorithm and its cybersecurity applications.
- DOI: <https://doi.org/10.5281/zenodo.21126582>

INTERNSHIP EXPERIENCE

Software Development Intern — Hope Foundation, Bangalore

Jan 2025 – Feb 2025

- Completed a 90-hour hands-on internship, building a Social Media Application from scratch.
- Delivered the project on schedule, demonstrating professional conduct and consistent commitment.

Deloitte Australia – Cyber Job Simulation (Forage)

July 2026

- Investigated a simulated cybersecurity incident using web activity logs.
- Identified suspicious user activity and analyzed a potential data breach.
- Applied log analysis, networking, and web security concepts in an industry-style scenario

CERTIFICATIONS

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- Python for Beginners — Great Learning Academy (Live Course)
 - Machine Learning & AI — Hope Foundation, Bangalore (2024)
 - Cybersecurity (Basic & Intermediate) — Simplilearn (2024)
 - Deloitte Australia Cyber Job Simulation – Forage (July 2026)

ACHIEVEMENTS

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- Participated in **Adobe University Hackathon 2026** organized by Adobe and Unstop.
 - Deployed a live AI application (**DocuMind AI**) on Streamlit Cloud.
 - Completed a **90-hour** professional internship at Hope Foundation.
 - Maintained a consistent academic record with **7.8** CGPA throughout BCA.

LANGUAGES

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- Kannada — Native
 - English — Professional Working Proficiency
 - Hindi — Conversational